Basic non-parametric tests

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July 22, 2026

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Concepts

- Non-parametric tests are the set of fundamental tests conducted when the data does not meet the assumption of the parametric tests
- So, non-parametric test can be used when:
    - The sample is too small ($n < 30$)
    - non-normally distributed data
- Similar to parametric tests, non-parametric test require a numerical data
- There are 3 most common basic non-parametric tests:
    1. Mann-Whitney test (equivalent to independent t-test)
    2. Wilcoxon-signed rank test (equivalent to paired t-test)
    3. Kruskal-Wallis test (equivalent to one-way ANOVA)

Mann-Whitney test

- Compares the numerical variable of the two independent groups
- Other names: Mann-Whitney U test, Wilcoxon-Mann-Whitney test, Wilcoxon-rank sum test
- Mann-Whitney test is conducted when:
  - The dependent variable is numerical
  - The independent/grouping variable is categorical (2 levels)
- Assumptions:
  1. Random sample
  2. The grouping and the outcome variables are independent of each other
- Examples:
  - Comparing the systolic blood pressure between patients clinic A and clinic B
  - Comparing the BMI of doctors and pharmacists in a hospital

Mann-Whitney test: Analysis

- Background: we want to compare the pain score between groups A and B
- Steps:
  1. State the hypothesis
  2. Check the assumptions
  3. Interpretation
  4. Conclusion

- Step 1:
  - H_0: There is no difference in pain score between groups A and B
  - H_A: There is a difference in pain score between groups A and B

```r
# Packages
library(foreign)
library(knitr)
library(ggplot2)
library(dplyr)

# Read the data
dat <- read.spss("Data/mw_data.sav", to.data.frame = TRUE)
```

```r
# First 6 rows of the data
head(dat) %>% kable()
```

Patient_ID	Group	Pain_Score
1	A	4
2	A	3
3	A	2
4	A	5
5	A	6
6	A	5

- Step 2: Assumptions:
  1. Check the theoretical assumptions
  2. In our data, $n = 20$ (so can directly do the test)
- Step 3: Interpret:
  - p value < 0.05
  - There is a significant difference in the pain score between groups A and B

```r
# Mann-Whitney test
wilcox.test(Pain_Score ~ Group, dat, exact = F)
```

```
    Wilcoxon rank sum test with continuity correction

data:  Pain_Score by Group
W = 5.5, p-value = 0.0007639
alternative hypothesis: true location shift is not equal to 0
```

- Step 4: Conclusion:
    - Group A had a significantly lower pain score compared to group B

```r
dat %>%
    group_by(Group) %>%
    summarise(median = median(Pain_Score))
```

```
# A tibble: 2 x 2
  Group median
  <fct>  <dbl>
1 A        3.5
2 B        6.5
```

Wilcoxon-signed rank test

- Compares the two dependent or related samples:
  - Pre and post
  - Matched study design
  - Closely related subjects like twins
  - Cross over study design
- Wilcoxon-signed rank test is conducted when:
  - The dependent variable is numerical
  - The groups are related
- Assumptions:
  1. Random sample
  2. Dependent/related sample

- Examples:
  - Comapring the cholesterol level before and after 6 month of physical exercise
  - Comparing the effective of two drugs (A and B) in weight loss (cross over study design)

Wilcoxon-signed rank test: Analysis

- Background: we want to compare the discomfort back pain score between pre and post stretching session level (a higher score indicates a more severe discomfort)
- Steps:
  1. State the hypothesis
  2. Check the assumptions
  3. Interpretation
  4. Conclusion

- Step 1:
  - H_0: There is no difference in discomfort score pre and post stretching session
  - H_A: There is a difference in discomfort score pre and post stretching session

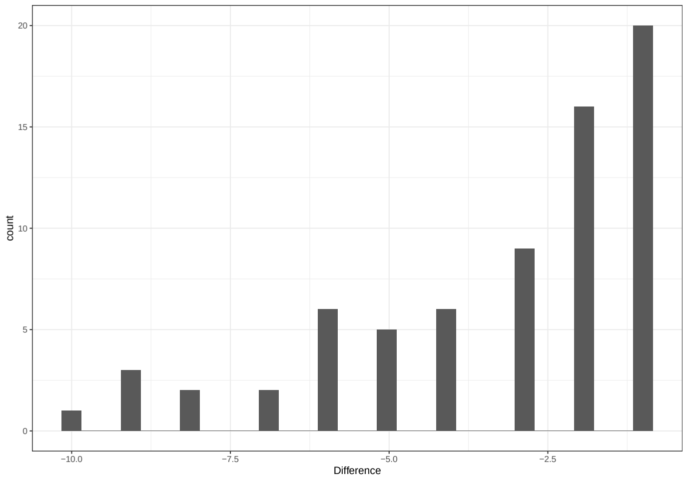
```r
# Packages
library(foreign)
library(knitr)
library(ggplot2)

# Read the data
dat <- read.spss("Data/wilcoxon_data.sav", to.data.frame = TRUE)
```

```r
# First 6 rows of the data
head(dat) %>% kable()
```

ID	Pre_Score	Post_Score	Difference
1	13	12	-1
2	18	9	-9
3	8	7	-1
4	10	8	-2
5	13	10	-3
6	32	31	-1

- Step 2:
  1. Check the theoretical assumptions
  2. In our data, n = 70 - data does not looks normal

```
ggplot(dat, aes(Difference)) +
    geom_histogram() +
    theme_bw()
```



count
20
15
10
5
0
−10.0
−7.5
−5.0
−2.5
Difference

- Step 3: Interpret:
  - p value $<$ 0.05
  - There is a significant difference in discomfort score pre and post stretching session

```r
# Wilcoxon-signed rank test
wilcox.test(dat$Pre_Score, dat$Post_Score, paired = T, exact = F)
#there is a ties or large sample exact = F should be used,
#exact = T is the best if no ties
```

- Step 4: Conclusion:
    - The stretching session reduce the discomfort among the participant (median post $<$ median pre)

```r
dat %>%
    summarise(
        median_pre = median(Pre_Score),
        median_post = median(Post_Score)
        )
```

```
  median_pre median_post
1         13          10
```

Kruskal-Wallis test

- Compares the numerical values of more than 2 groups
- Other names: one-way ANOVA on ranks test. Kruskal-Wallis H test
- Kruskal-Wallis test is conducted when:
  - The dependent variable is numerical
  - The independent/grouping variable is categorical ($>$2 levels)
- Assumptions:
  1. Random sample
  2. The grouping and the outcome variables are independent of each other

- Examples:
  - Comparing the systolic blood pressure between patients clinic A, B, and C
  - Comparing the BMI of doctors, pharmacists, and dentists in a hospital
- Similar to one-way ANOVA, Krukal-Wallis test requires a post-hoc test if it is significant
- So:
  - Krukal-Wallis test: is there significant difference between the groups?
  - Post-hoc test: which group pairwise comparison is significant?
- Post-hoc test consists of doing several Mann-Whitney tests for each pairwise comparison
- Similar to the post-hoc test in one-way ANOVA, since the post-hoc tests are conducted several times on the same dataset, bonferroni adjustment is needed to adjust the p value (to avoid false positive)

Kruskal-Wallis test: Analysis

- Background:
  - We want to compare the different teaching style used in 3 different classes (classes A, B, and C)
  - At the end of the semester, the students were given the quiz to assess the effectiveness of each teaching style
- Steps:
  1. State the hypothesis
  2. Check the assumptions
  3. Interpretation
  4. Conclusion

- Step 1:
  - H_0: There is no difference in quiz score between classes A, B, and C
  - H_A: There is a difference in quiz score for at least one of the classes

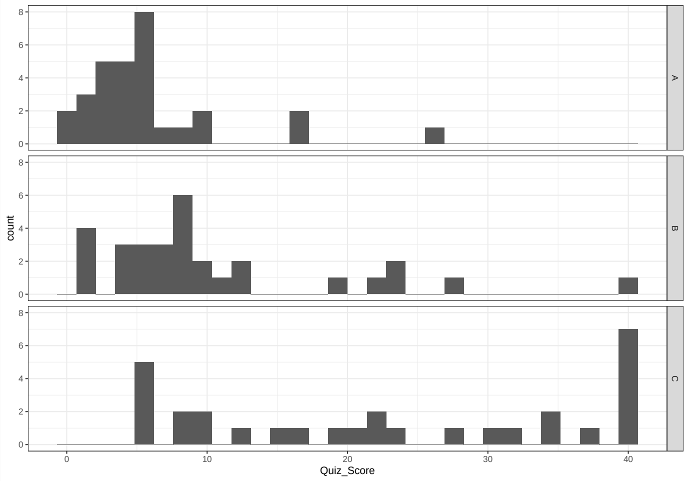
```r
# Packages
library(foreign)
library(knitr)
library(ggplot2)
library(dplyr)

# Read the data
dat <- read.spss("Data/kw_data.sav", to.data.frame = TRUE)
```

```r
# First 6 rows of the data
head(dat) %>% kable()
```

ID	Class	Quiz_Score
1	A	6
2	A	6
3	A	2
4	A	5
5	A	3
6	A	8

- Step 2:
    1. Check the theoretical assumptions
    2. In our data, n = 90 (each group n = 30)

```r
ggplot(dat, aes(Quiz_Score)) +
    geom_histogram() +
    facet_grid(rows = vars(Class)) +
    theme_bw()
```



count
Quiz_Score
A
B
C
0
2
4
6
8
0
10
20
30
40

- Step 3: Interpret Kruskal-Wallis test:
    - p value < 0.05
    - There is a difference in the quiz score for at least one of the classes
    - So, post-hoc test is indicated

```r
# Kruskal-Wallis test
kruskal.test(Quiz_Score ~ Class, dat)
```

```
	Kruskal-Wallis rank sum test

data:  Quiz_Score by Class
Kruskal-Wallis chi-squared = 33.775, df = 2, p-value = 4.632e-08
```

- Step 3: Interpret post-hoc test:
    - There is a significant difference between all pairwise comparison among the classes

```r
pairwise.wilcox.test(
    dat$Quiz_Score, dat$Class,
    p.adjust.method = "bonferroni"
    )
```

```
    Pairwise comparisons using Wilcoxon rank sum exact test

data:  dat$Quiz_Score and dat$Class

  A       B
B 0.01541 -
C 5.1e-09 0.00062

P value adjustment method: bonferroni
```

- Step 4: Conclusion:
  - There is a significant difference between the quiz score among the classes
  - Teaching used in class C was the most effective, followed by the one in class B and A, respectively

```
dat %>%
    group_by(Class) %>%
    summarise(median = median(Quiz_Score))
```

```
# A tibble: 3 x 2
  Class median
  <fct>  <dbl>
1 A        4.5
2 B        8
3 C       22
```